

L01 — Pre-class Reading

The cold brick, and the idea the whole course stands on

This is the print edition of the interactive pre-class page. Ten short steps, each deeper than the last, rebuilt from the September 9 lecture: one brick parked in dry ice quietly contains the entire first half of PHYS 157 — and the one idea underneath all of it: **we never track 10^{23} particles; we track their averages.**

Instructor Dr. Simin Najmi · Section 102 | **Lecture date** September 9, 2026

Companion interactive page at finneaswang.github.io/phys-course-os | **Obsidian vault** PHYS 157

Sources L1-Introduction.pdf · L1-Intro-post.pdf (canonical) · Wednesday.txt (calibrated transcript)

Review copy for content QA — physics accuracy, ordering, question quality. Print: A4, vector text.

How this works, and the two pillars

This page is your **pre-class reading** for L01 — the part to master *before* re-watching the lecture or the quiz. Ten short steps, each one deeper than the last.

By the end you can:

- tell the cold-brick story **twice** — macroscopically and microscopically;
- distinguish **temperature**, **heat**, and **thermal equilibrium**;
- explain why thermodynamics is possible at all;
- say what L01 deliberately does **not** contain yet.

Pillar 1 — Thermodynamics (first half of term)

Macroscopic variables (T, V, p), thermal expansion, phase change, heat and its transfer methods, equilibrium. Built on averages over $\sim 10^{23}$ particles.

Pillar 2 — Oscillations & Waves (second half)

Collective motion of the same molecules: guitar strings, sound, ocean waves. Described by a wave equation — again, no per-particle tracking.

⚠ L01 contains **no formulas**. If you catch yourself deriving one here, you're ahead of the lecture — park it.

The cold brick — predict first

September 9, Hebb building. Prof pulls a brick out of a cooler. The room's first guess: *liquid nitrogen*. Wrong — it sat in **dry ice**, so the brick starts at about **$-78.5\text{ }^{\circ}\text{C}$** . Then it just sits in the $\sim 20\text{ }^{\circ}\text{C}$ room while everyone talks about it.

Predict before you read on — *in the first minute, what do you actually see?*

X A · Nothing — the brick just sits there, unchanged. Look at the temperature difference: $-78.5\text{ }^{\circ}\text{C}$ vs $\sim 20\text{ }^{\circ}\text{C}$. Something dramatic happens at the surface — and it is not silence.

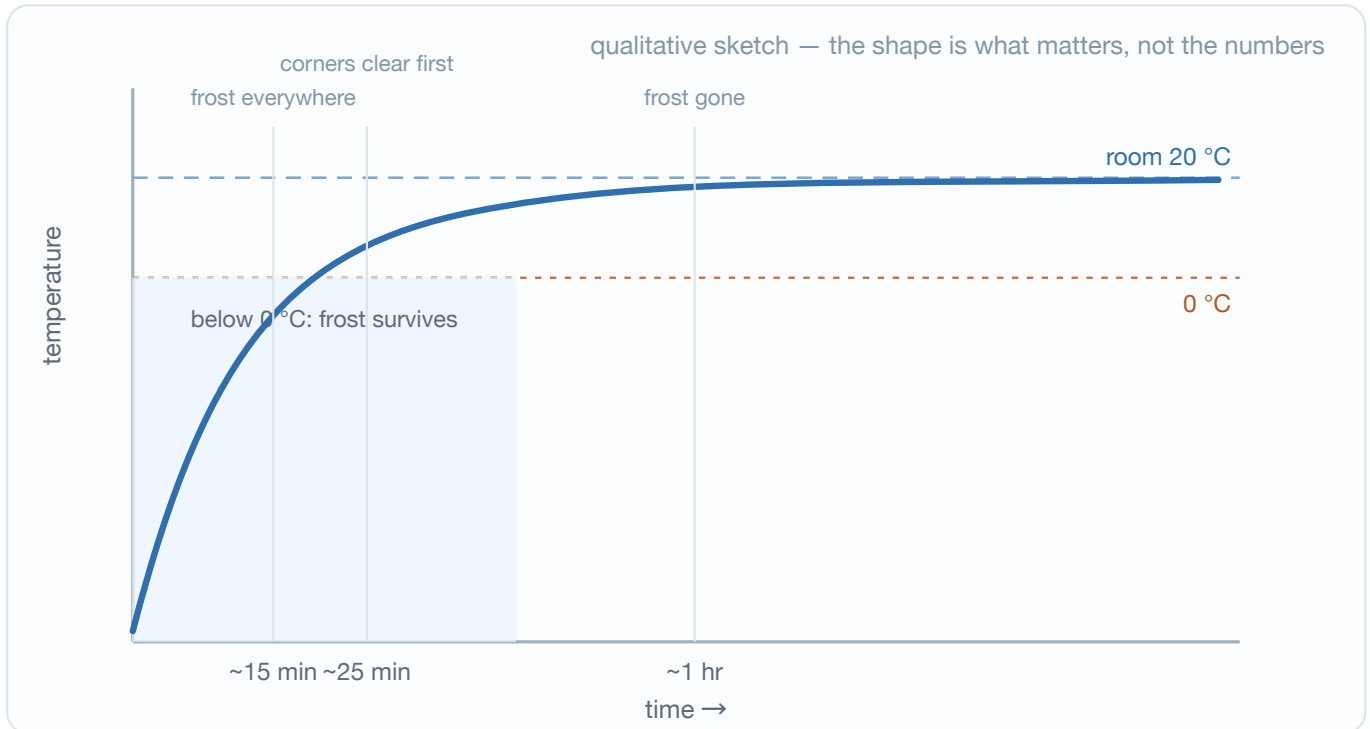
✓ B · Frost forms on the brick's surface. Right. Water vapour in the air touches a surface far below $0\text{ }^{\circ}\text{C}$ and leaves the gas phase $\rightarrow$ frost forms almost immediately. Meanwhile heat starts flowing into the brick and it begins warming.

X C · The brick glows faintly. Glowing usually means visible light — that needs a far higher temperature. The brick IS emitting thermal radiation right now, mainly in the infrared, which your eyes cannot detect.

X D · Water pools underneath it. Meltwater only appears once the surface climbs past $0\text{ }^{\circ}\text{C}$ — not in the first minute. First comes frost.

What actually happened (slide photos)

- **~15 min:** frost over the whole brick.
- **~25 min:** frost retreating — **corners and edges clear first** (most exposed surface per amount of brick → they warm fastest).
- **~1 hr:** brick nearly clear, temperature approaching room temperature.



The brick's temperature over its hour on the desk: it approaches room temperature — fast at first, then slower. Frost can only exist while the surface is below the 0 °C line.

Wording trap: slides say *condensation*, but at -78.5 °C the vapour goes from gas **straight to solid** — deposition (凝华). The photo is unmistakably frost, not liquid. For the quiz, answer with the instructor's wording; keep "deposition" as your private precision.

PRE-CLASS · STEP 3 / 10

The macroscopic view

Macroscopic = the scale you can see or measure with instruments. In that language, here is everything the brick did — five measurable statements:

- **Heat flowed in** — from everything warmer: the air, the desk it sat on. Big temperature difference → fast flow at the start.
- **Its temperature rose** — continuously, then more slowly.
- **Its dimensions grew** — every linear dimension, slightly. (Thermal expansion: first real topic of the course.)
- **Frost formed, then cleared** — a phase change happening in the air next to the brick.
- **It stopped changing** — brick and room reached the same temperature. That endpoint is thermal equilibrium.

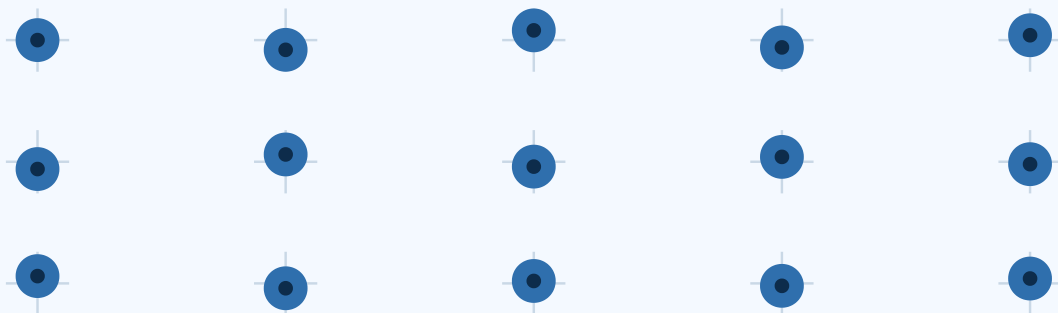
Notice what this language never mentions: atoms. That's the point — macroscopic variables are chosen so you *don't need* the atoms. Next step: the same story in the other language.

PRE-CLASS · STEP 4 / 10

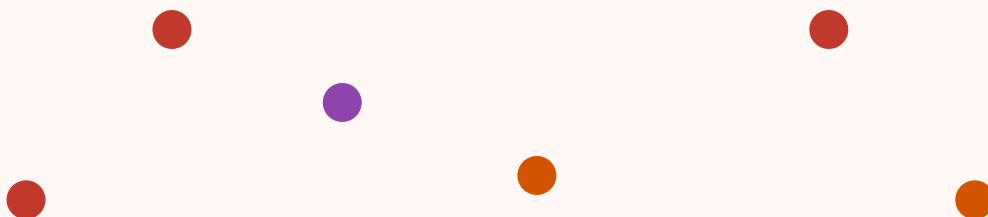
The microscopic view

Same brick, atomic vocabulary — three processes:

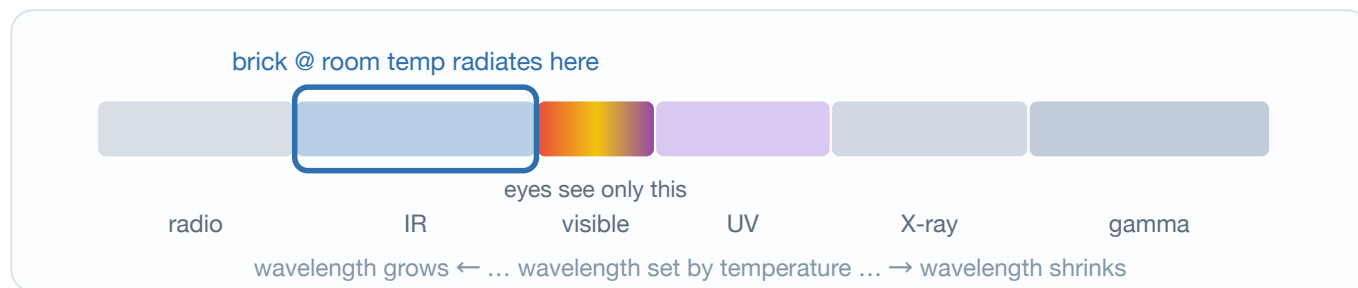
- **The brick's atoms vibrate.** It's a solid: atoms are not free to wander — they **vibrate about fixed equilibrium positions**. Cold brick = slow, low-energy vibrations. (This contrast is quiz-bait.)
- **Air molecules collide with it.** Fast molecules strike the brick's surface; each impact transfers a bit of momentum and energy. Billions of tiny deliveries per second = "heat flowing in".
- **Everything radiates.** Any object, any temperature, emits electromagnetic waves; the wavelength is set by temperature. Room temperature → mainly **infrared** — real energy transfer, invisible to your eyes. IR-camera demo with two warm bricks promised for next lecture.



SOLID — each atom vibrates about its own equilibrium site (cross = that site)



GAS — molecules travel long distances between collisions



The electromagnetic spectrum. The brick at room temperature radiates in the infrared band — just left of the tiny slice your eyes can see.

Go deeper — is energy flowing one way, molecule by molecule? No — and hold onto this. **Individual collisions can transfer energy in either direction.** Because the warmer side transfers more energy on average, there is a **net energy flow from hot to cold** — billions of tiny deliveries adding up. The word doing the work is **net**. It comes back at equilibrium.

PRE-CLASS · STEP 5 / 10

The bridge: averages become variables

Why can two such different stories describe one brick? Because of the idea the whole course stands on:

microscopic behaviour → statistical averages → macroscopic variables → thermodynamics

A brick holds $\sim 10^{23}$ particles. Tracking each is impossible — and, beautifully, **unnecessary**, because a few averages summarize everything we care about:

- The brick's overall **size** (Volume) reports the **average spacing** between atoms — which is exactly why thermal expansion is visible at all.
- The brick's **temperature** (Temperature) is the **average kinetic energy** (Kinetic Energy) of its vibrating particles. Prof's phrasing, worth memorizing: *"the average of all those kinetic-energy values is what we call temperature."*

That is what **Thermodynamics** is: the art of summarizing 10^{23} things into a handful of variables, then finding the rules those summaries obey. Measuring the macroscopic scale tells you what the microscopic world is doing **on average**.

PRE-CLASS · STEP 6 / 10

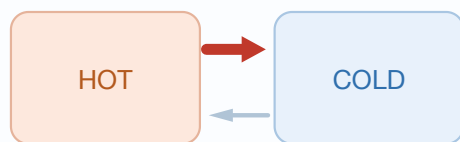
Temperature vs Heat vs Equilibrium

Three words the quizzes love to blur. Three clean definitions:

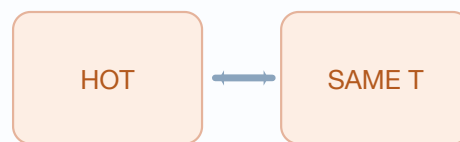
Temperature — a **property of the object**, a *state variable* (状态量). It describes what the system *is* — in the L01 model, the average kinetic energy of its particles.

Heat — **energy in transit**, driven by a temperature difference. Not a property of anything — it only exists *while crossing a boundary*, carried by collisions and radiation.

Thermal Equilibrium — the endpoint of heat flow: common temperature, and the **net** exchange drops to zero because the two averages have matched.



thick = net flow; thin = some returns



equal exchange: net = 0, arrows never vanish

equilibrium = balanced traffic, not an empty road

Heat traffic: before equilibrium (left) the net flow is one-way; at equilibrium (right) the exchange is equal in both directions — the arrows never disappear, they only cancel.

The two traps

- **"The brick contains heat" — wrong.** An object contains *internal energy*; "heat" names energy *while it moves between things* — the way "flow" describes water in a pipe, not water sitting in a bucket.
- **"At equilibrium everything stops" — wrong.** Collisions continue, energy crosses both ways every instant. Equilibrium is **balanced traffic, not an empty road**.

PRE-CLASS · STEP 7 / 10

The waves teaser

Last five minutes of L01, and easy to dismiss — don't. The brick was also being hit by **sound** from every voice in the room and by light from the ceiling. Those are **waves**:

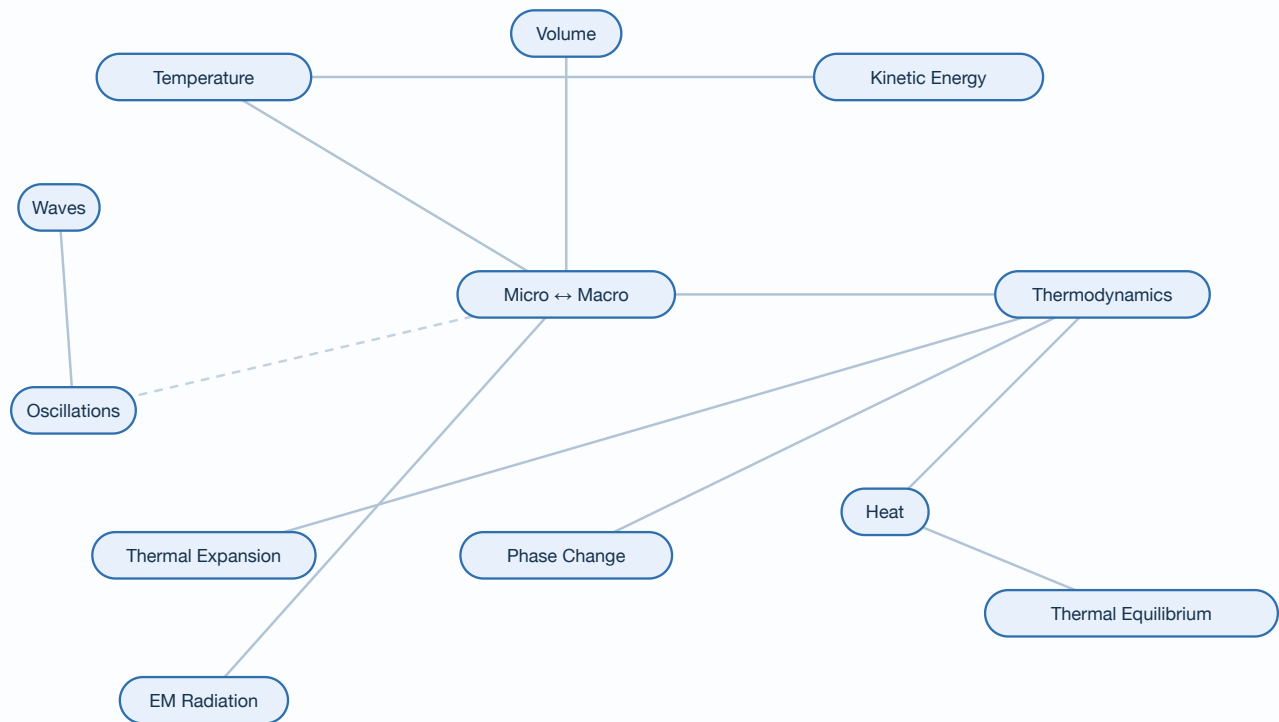
- a wave is **collective motion** — enormous numbers of particles moving in coordination, so that **energy travels without the material travelling**;
- examples named in lecture: sound waves, guitar strings, ocean waves; oscillations can be **transverse** or **longitudinal**;
- and the same trick as thermodynamics applies: nobody will track 10^{23} particles — a **wave equation** summarizes the collective behaviour.

Both pillars, one idea: *stop tracking individuals; summarize*. Thermodynamics does it with averages; waves do it with a wave equation. That parallel is the deepest thing L01 plants.

PRE-CLASS · STEP 8 / 10

Your concept graph

This is the network your course is building. Every node is a concept page (in the interactive version, clicking opens it). The dashed link is the second pillar — preview-only for now.



• every node = a concept page · dashed = previewed, not yet taught · this graph grows with every lecture.

PRE-CLASS · STEP 9 / 10

Check yourself

Six conceptual questions. Answer out loud *first*, then read the answer. If you can't say it, you don't have it yet — go back one step.

1 · Why is tracking every particle impossible — and why is that fine?

A brick has $\sim 10^{23}$ particles; no one can follow them. It's fine because the **averages** are all that matter, and averages appear as macroscopic variables (size $\leftrightarrow$ spacing, temperature $\leftrightarrow$ average kinetic energy). Summaries + rules = thermodynamics.

2 · How does a brick atom's motion differ from an air molecule's?

Brick atoms are a solid: they **vibrate about fixed positions**. Gas molecules **travel long distances** between collisions. "Vibrate" vs "travel" — don't blend the pictures.

3 · Which is the state variable — temperature or heat? Define the other.

Temperature is the state variable (it describes what the object *is*). Heat is **energy in transit**, driven by a temperature difference, carried by collisions and radiation. Objects contain internal energy, never "heat".

4 · At thermal equilibrium, what exactly stops — and what continues?

The **net** energy flow stops (temperatures equal, averages match). Individual collisions continue in **both** directions forever. Balanced traffic, not an empty road.

5 · Why did the frost clear from corners and edges first?

Corners/edges have the most **exposed surface per amount of brick**, so they collect heat fastest and warm first. Geometry controls heat-flow rate.

6 · The brick radiates constantly — why can't you see it?

At room temperature the emission is mainly **infrared**: wavelength set by temperature, and the visible band is a narrow slice our eyes evolved for. The radiation is real and carries energy — just not in colors we detect.

PRE-CLASS · STEP 10 / 10

60-second review — and done

The whole pre-class, one breath

- Macroscopic variables = **averages** over $\sim 10^{23}$ particles; thermodynamics studies the rules of those averages.
- **Temperature** = average kinetic energy of particles *in the L01 model* — a state variable.
- **Heat** = energy in transit, hot $\rightarrow$ cold, carried by **collisions + radiation**; never "stored".
- Cold brick: heat in $\rightarrow$ warms $\rightarrow$ expands; frost **deposits**, clears **corners first**, gone by ~ 1 hr.
- **Equilibrium**: same temperature, **net** flow zero, two-way collisions continue.
- Everything radiates; wavelength set by temperature; room-T = infrared, invisible.
- Waves = **collective motion** — second pillar, same "summarize, don't track" trick.

Every blue term in the interactive version opens one of these

INTRODUCED · L01

Microscopic & Macroscopic Description 微观与宏观描述

Two vocabularies for the **same** object: microscopic — what atoms/molecules do (vibrate about fixed sites in solids; translate and collide in gases; radiate); macroscopic — the few measurable variables encoding the *averages*. Chain: microscopic behaviour → statistical averages → macroscopic variables → thermodynamic laws. Mappings: size ↔ spacing; T ↔ average KE.

INTRODUCED · L01

Temperature 温度

A macroscopic variable and a **state variable**: in the model introduced in L01, temperature reflects the **average kinetic energy** of a system's particles. Sets the direction of spontaneous heat flow (hot → cold) and the endpoint (equilibrium).

INTRODUCED · L01

Volume 体积

Macroscopic variable reporting the **average spacing** between atoms/molecules — why thermal expansion is visible at all.

INTRODUCED · L01

Kinetic Energy 动能

Energy of motion, $E_k = \frac{1}{2}mv^2$. Thermal role: temperature reflects the *average* kinetic energy of particles — in the brick, of atoms vibrating about their sites.

INTRODUCED · L01

Thermodynamics 热力学

The study of how macroscopic variables summarize $\sim 10^{23}$ particles and the rules those summaries obey. You never track individuals — you measure a handful of averages, and those averages obey rules.

INTRODUCED · L01

Heat 热

Energy in transit driven by ΔT — "the flow of thermal energy from one object to the other". Carriers: molecular collisions; radiation (infrared at room temperature). ⚠ An object contains internal energy — it does **not** contain heat.

INTRODUCED · L01

Thermal Equilibrium 热平衡

Endpoint of heat flow: common temperature; **net** exchange = 0 while two-way collisions continue. Balanced traffic, not an empty road.

PREVIEWED · L01 **Phase Change** 相变

Solid/liquid/gas transitions. L01: vapour → frost on the $-78.5\text{ }^{\circ}\text{C}$ brick — strictly **deposition** (凝华), though the slides' shorthand says "condensation". Latent heat comes later.

PREVIEWED · L01 **Thermal Expansion** 热膨胀

Every linear dimension grows with temperature (microscopically: average spacing grows). Formal quantitative law arrives in the next lectures.

PREVIEWED · L01 **Oscillations** 振动

Repeated motion about an equilibrium position — already true inside the brick (atoms vibrate about lattice sites). Second half builds single oscillators → waves.

PREVIEWED · L01 **Waves** 波

Travelling **collective motion**: energy handed from neighbour to neighbour without the material travelling. Wave equation later — no per-particle tracking.

INTRODUCED · L01 **Electromagnetic Radiation** 电磁辐射

Every object emits EM waves at any temperature; wavelength set by temperature; room-T → infrared (invisible). One of the energy-transfer channels. IR-camera demo promised for next lecture.